



Aerospace Control Systems

Exam project AA 25/26

Robust multivariable control of the lateral/longitudinal dynamics of a multirotor UAV

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Introduction

01

Robust multivariable control of the lateral/longitudinal dynamics of a multirotor UAV

- The ANT-X quadrotor is a small quadrotor drone designed for research and education.
- Grey-box models of ANT-X have been identified from flight test data and are available for control law design.
- The project task is to design and verify a robust multivariable attitude control system for the ANT-X quadrotor.



Robust multivariable control of the lateral/longitudinal dynamics of a multirotor UAV

- This presentation provides the mathematical models for the dynamics of the longitudinal and lateral axes of the drone and a detailed presentation of the project tasks.
- Numerical values for the parameters of the model are included, as well as the corresponding uncertainties.





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Linearised models of the lateral and longitudinal dynamics

02

Linearised model for lateral dynamics

The linearised lateral dynamics of the drone are given by:

$$\begin{aligned}\dot{x} &= A_{lat}x + B_{lat}u_{lat} \\ y &= Cx + Du_{lat}\end{aligned}$$

where

$$\begin{aligned}u_{lat} &= \delta_{lat}, \quad y = \varphi \\ x &= \begin{bmatrix} v \\ p \\ \varphi \end{bmatrix}\end{aligned}$$

and

$$A_{lat} = \begin{bmatrix} Y_v & Y_p & g \\ 0 & L_p & 0 \\ 0 & 1 & 0 \end{bmatrix}, B_{lat} = \begin{bmatrix} Y_\delta \\ L_\delta \\ 0 \end{bmatrix}, C = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, D = 0$$

Linearised model for lateral dynamics

The control variable

$$u_{lat} = \delta_{lat}$$

is a normalised pitching moment ranging between -1 and 1.

The state vector

$$x = \begin{bmatrix} v \\ p \\ \varphi \end{bmatrix}$$

includes lateral velocity v , roll rate p and roll angle φ .

Lateral dynamics: parameters and uncertainties

The numerical values of the parameters are given by ($g = 9.81$)

Stability derivatives

$$Y_v = -0.1068 \text{ 1/s (4\%)}$$

$$Y_p = -0.1192 \text{ m/s rad (2\%)}$$

$$L_p = 2.6478 \text{ 1/s (2\%)}$$

Control derivatives

$$Y_d = 10.1647 \text{ m/s}^2 \text{ (1.5\%)}$$

$$L_d = 450.7085 \text{ rad/s}^2 \text{ (1\%)}$$

where uncertainty is given in terms of standard deviations (provided as percentage of corresponding nominal values) assuming a Gaussian density for each parameter.

Linearised model for longitudinal dynamics

The linearised longitudinal dynamics of the drone are given by:

$$\begin{aligned}\dot{x} &= A_{lon}x + B_{lon}u_{lon} \\ y &= Cx + Du_{lon}\end{aligned}$$

where

$$\begin{aligned}u_{lon} &= \delta_{lon}, \quad y = \theta \\ x &= \begin{bmatrix} u \\ q \\ \theta \end{bmatrix}\end{aligned}$$

and

$$A_{lon} = \begin{bmatrix} X_u & X_q & -g \\ 0 & M_q & 0 \\ 0 & 1 & 0 \end{bmatrix}, B_{lon} = \begin{bmatrix} X_\delta \\ M_\delta \\ 0 \end{bmatrix}, C = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, D = 0$$

Linearised model for longitudinal dynamics

The control variable

$$u_{lon} = \delta_{lon}$$

is a normalised pitching moment ranging between -1 and 1.

The state vector

$$x = \begin{bmatrix} u \\ q \\ \theta \end{bmatrix}$$

includes longitudinal velocity u , pitch rate q and pitch angle θ .

Longitudinal dynamics: parameters and uncertainties

The numerical values of the parameters are given by ($g = 9.81$)

Stability derivatives

$$X_u = -0.1068 \text{ 1/s (4\%)}$$

$$X_q = 0.1192 \text{ m/s rad (2\%)}$$

$$M_q = -2.6478 \text{ 1/s (2\%)}$$

Control derivatives

$$X_d = -10.1647 \text{ m/s}^2 \text{ (1.5\%)}$$

$$M_d = 450.7085 \text{ rad/s}^2 \text{ (1\%)}$$

where uncertainty is given in terms of standard deviations (provided as percentage of corresponding nominal values) assuming a Gaussian density for each parameter.

Longitudinal/lateral coupling

While the nominal longitudinal/lateral dynamics are uncoupled, an uncertain coupling must be considered:

$$\begin{bmatrix} u_{lon} \\ u_{lat} \end{bmatrix} = \begin{bmatrix} \cos(\psi) & \sin(\psi) \\ -\sin(\psi) & \cos(\psi) \end{bmatrix} \begin{bmatrix} \delta_{lon} \\ \delta_{lat} \end{bmatrix}$$

where ψ ranges between -15° and $+15^\circ$.



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Description of the project tasks

03

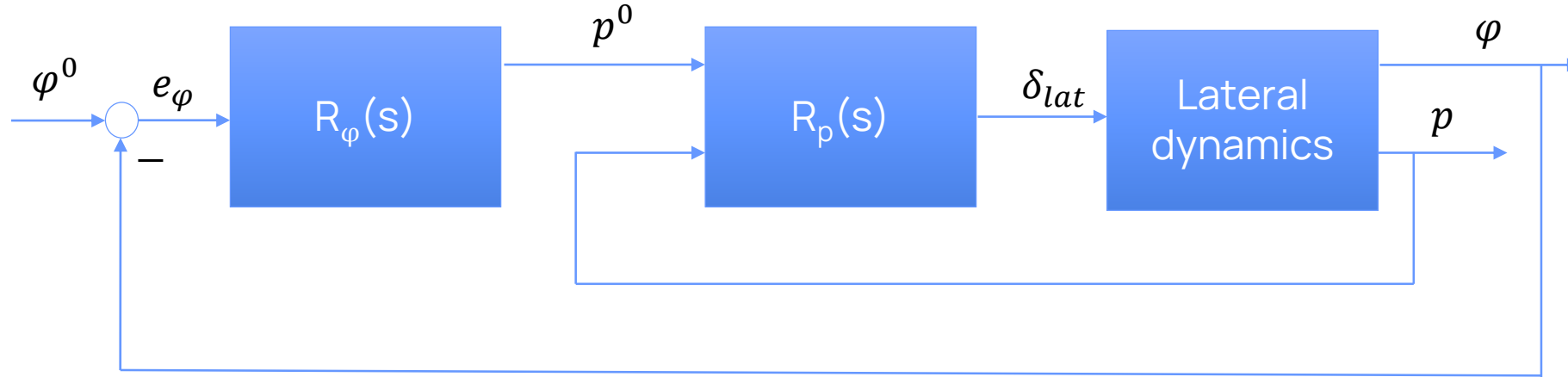
Task 1: Nominal modelling and design

Task 1: Nominal modelling and design

- Build design models for the quadrotor dynamics using the available nominal values.
- Analyse the models in terms of poles, zeros, frequency response functions.
- Design the attitude control loops using the models and nominal values for the parameters, based on the architecture and on the requirements in the next slides.

Task 1: Nominal modelling and design

The lateral attitude control system is represented in the block diagram.



The roll rate ($R_p(s)$) and roll angle ($R_\varphi(s)$) controllers are, respectively, a PID and a P controller.

Task 1: Nominal modelling and design

Controller $R_p(s)$:

$$\delta_{lat} = K_{p,P}e_p + \frac{K_{p,I}}{s}e_p + \frac{K_{p,D}s}{1+sT_p}(-p), T_p=0,02\text{ s},$$
$$e_p = p^0 - p$$

Controller $R_\varphi(s)$:

$$p^0 = K_\varphi e_\varphi,$$
$$e_\varphi = \varphi^0 - \varphi$$

Task 1: Nominal modelling and design

Task 1 design requirements:

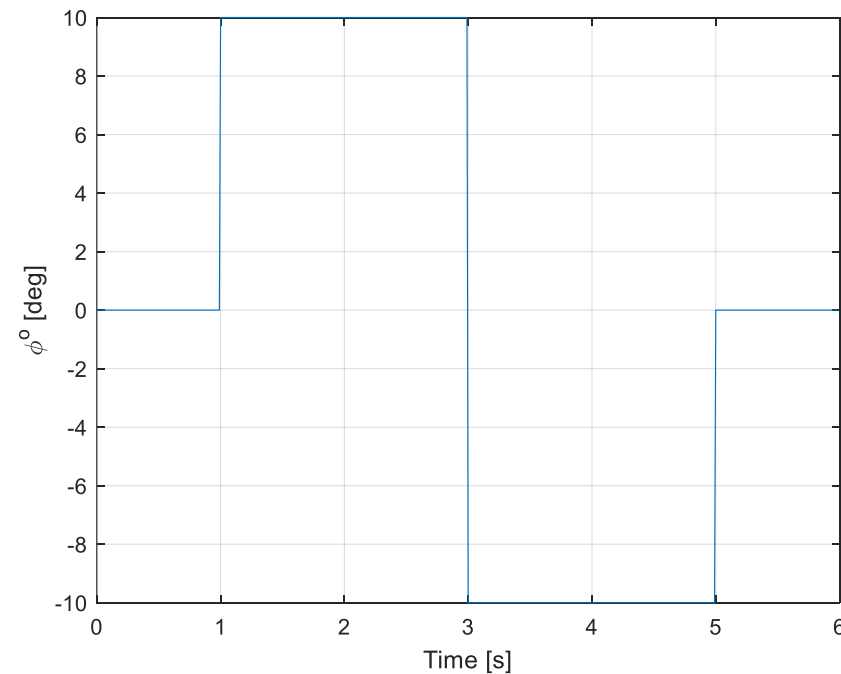
A. Nominal Performance:

Response of φ to variations in φ^0 : equivalent to a second-order response with $\omega_n \geq 10 \text{ rad/s}$ and $\xi \geq 0.9$.

Task 1: Nominal modelling and design

Task 1 design requirements:

- B. Control effort limitation: $|\delta_{lat}| \leq 0.05$ for the doublet change in φ^0 (with $10 \pi/180$ rad amplitude) given by (in deg)



Task 1: Nominal modelling and design

Task 1 design requirements:

The architecture and the requirements for the longitudinal axis are identical to the ones for the lateral axis.

Task 2: Uncertain modelling

Task 2: Uncertain modelling

using $\pm 3\sigma$ ranges as deterministic bounds for parametric uncertainty:

- Build a model for the quadrotor dynamics using the available information and the MATLAB Robust Control Toolbox for uncertain system representation.
- Build the uncertain linearised models for lateral dynamics and for the longitudinal dynamics using the available information and the MATLAB Robust Control Toolbox for uncertain system representation.
- Analyse the models in terms of poles, zeros, frequency response functions.

Task 3: Robustness analysis

Task 3: Robustness analysis:

- Express the uncertain models in $P - \Delta$ form, in terms of the uncertain parameters.
- Verify that the obtained control systems satisfies the RS condition for the uncertain models in $P - \Delta$ form.
- If not, redesign to achieve RS.

(the use of MATLAB functions `robstab`, `robgain`, `robustperf` is not allowed)

Task 4: Robustness to coupling

Task 4: Robustness to coupling

- Verify if the obtained control system satisfies the RS condition when taking into account the longitudinal/lateral coupling given by

$$\begin{bmatrix} u_{lon} \\ u_{lat} \end{bmatrix} = \begin{bmatrix} \cos(\psi) & \sin(\psi) \\ -\sin(\psi) & \cos(\psi) \end{bmatrix} \begin{bmatrix} \delta_{lon} \\ \delta_{lat} \end{bmatrix}$$

where ψ ranges between -15° and $+15^\circ$.

(the use of MATLAB functions `robstab`, `robgain`, `robustperf` is not allowed)

Task 5: Control system validation

Task 5: Control system validation

- Build a complete model for the quadrotor dynamics using the available information and the MATLAB Robust Control Toolbox for uncertain system representation, using:
 - Gaussian densities for the uncertain parameters
 - Uniform density for ψ .
- Implement Monte Carlo simulation on the complete uncertain feedback system and analyse compliance with the design requirements justifying the choice of the number of MC iterations.



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Description of the deliverables

04

Description of the deliverables

When the tasks are complete prepare a presentation (Powerpoint preferred) including

- An outline of the considered modelling approach
- An outline of the considered design approach
- An overview of the developed Matlab code
- A presentation of the results in terms of the given task structure.

Submit a .zip file with the MATLAB code developed to execute the project.



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Evaluation process

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Slides submission

Presentation slides must be submitted at least **48 hours before** the oral discussion.

Advance submission:

- supports preliminary technical assessment,
- enables targeted oral discussion,
- improves consistency and fairness of evaluation.

MATLAB code submission

MATLAB code must be submitted together with the presentation material.

Code is not evaluated in terms of programming style.

However, it may be inspected:

- to verify methodological consistency,
- to clarify implementation details,
- to assess effective ownership of the analysis.

Evaluation focuses on correctness, traceability and ownership.

Use of AI-based tools

AI-based systems may be used as support tools during project development.

However:

- students remain fully responsible for all technical content,
- modelling, analysis and design choices must be independently understandable and defensible,
- inability to justify submitted material negatively affects the evaluation.

Evaluation targets demonstrated competence and methodological ownership.

Evaluation approach

Projects are evaluated according to:

- Formal rigor
- Conceptual understanding
- Engineering reasoning
- Methodological ownership

Students must be able to independently discuss and justify submitted material.



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